
Vulnerability Assessment & Scanning Report

Internship Task: Task 2 - Network Security & Scanning

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Target IP: 192.168.56.101 (Metasploitable2)

ID:APSPL2631294

Date: April 24, 2026

1. Executive Summary

A comprehensive security assessment was conducted on the target host **192.168.56.101**. The host is in an extremely insecure state, with several critical backdoors active. These vulnerabilities allow an unauthenticated attacker to gain full administrative (root) access to the system remotely.

Vulnerability Distribution

Severity Level	Count	CVSS Score Range
Critical	4	9.0 - 10.0
High	6	7.0 - 8.9
Medium	12	4.0 - 6.9
Low	5	0.0 - 3.9

```
File Actions Edit View Help
kali@kali: ~
└─$ sudo gvm-start
[+] Please wait for the GVM services to start.
[+] You might need to refresh your browser once it opens.
[+] Web UI (Greenbone Security Assistant): https://127.0.0.1:9392

• gsd.service - Greenbone Security Assistant daemon (gsd)
  Loaded: loaded (/usr/lib/systemd/system/gsd.service; disabled; preset: disabled)
  Active: active (running) since Thu 2026-04-23 07:38:34 EDT; 80ms ago
  Invocation: 2fc73d48c7b249e2a30e0e26581f9b56
  Docs: man:gsd(8)
        https://www.greenbone.net
  Main PID: 63033 (gsd)
  Tasks: 1 (limit: 2258)
  Memory: 1.4M (peak: 2.1M)
  CPU: 23ms
  CGroup: /system.slice/gsd.service
          └─63033 /usr/sbin/gsd --foreground --listen 127.0.0.1 --port 9392

Apr 23 07:38:34 kali systemd[1]: Starting gsd.service - Greenbone Security Assistant daemon (gsd)...
Apr 23 07:38:34 kali systemd[1]: Started gsd.service - Greenbone Security Assistant daemon (gsd).
Apr 23 07:38:35 kali gsd[63033]: gsd main: INFO:2026-04-23 11h38.35 utc:63033: Starting GSAD version 24.16.0-git
Apr 23 07:38:35 kali gsd[63046]: gsd main: INFO:2026-04-23 11h38.35 utc:63046: Starting HTTP server on 127.0.0.1 and port 9392
Apr 23 07:38:35 kali gsd[63033]: gsd main: INFO:2026-04-23 11h38.35 utc:63033: Starting HTTPS server on 127.0.0.1 and port 9392
Apr 23 07:38:35 kali gsd[63046]: gsd main:CRITICAL:2026-04-23 11h38.35 utc:63046: Starting gsd redirect daemon failed!
Apr 23 07:38:35 kali gsd[63033]: gsd main: INFO:2026-04-23 11h38.35 utc:63033: gsd started successfully

• gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
  Loaded: loaded (/usr/lib/systemd/system/gvmd.service; disabled; preset: disabled)
```

2. Methodology

The assessment was performed using an integrated approach:

1. **Reconnaissance:** Port discovery across the full range (\$1 - 65535\$).
2. **Vulnerability Identification:** Use of Nmap Scripting Engine (NSE) and manual CVE cross-referencing.
3. **Analysis:** Categorization of risks based on the CVSS (Common Vulnerability Scoring System) standards.

```
File Actions Edit View Help
Shell No. 1
<get_tasks_response status="200" status_text="OK"><apply_overrides>0</apply_overrides><task id="c637f5d6-73df-4fed-8864-1af50ce672a0"><owner><name>admin</name></owner><name>Task 2 Vulnerability Assessment</name><comment></comment><creation_time>2026-04-24T03:44:44Z</creation_time><modification_time>2026-04-24T03:44:44Z</modification_time><writable>1</writable><in_use>0</in_use><permissions><permission><name>Everything</name></permissions></permissions><alterable>0</alterable><usage_type>scan</usage_type><config id=""><name></name></config><target id="b91b211c-88a3-4ca9-9316-13c30880c4ae"><name>Metasploitable2_Target</name></target><scanner id="6acd0832-df90-11e4-b9d5-28d24461215b"><name>CVE</name><type>3</type></scanner><status>New</status><progress>-1</progress><report_count>0</finished>0</report_count><trend></trend><schedule id=""><name></name></schedule><schedule_periods>0</schedule_periods><observers></observers><average_duration>0</average_duration><result_count>0</result_count><preferences><preference><name>Maximum concurrently executed NVTs per host</name><scanner_name>max_checks</scanner_name><value>4</value></preference><preference><name>Maximum concurrently scanned hosts</name><scanner_name>max_hosts</scanner_name><value>20</value></preference><preference><name>Add results to Asset Management</name><scanner_name>in_assets</scanner_name><value>no</value></preference><preference><name>Apply Overrides when adding Assets</name><scanner_name>assets_apply_overrides</scanner_name><value>yes</value></preference><preference><name>Min QOD when adding Assets</name><scanner_name>assets_min_qod</scanner_name><value>71</value></preference><preference><name>Auto Delete Reports</name><scanner_name>auto_delete</scanner_name><value>no</value></preference><preference><name>Auto Delete Reports Data</name><scanner_name>auto_delete_data</scanner_name><value>5</value></preference></preferences></task><filters id=""><term>apply_overrides=0 min_qod=70 first=1 rows=1000 uuid=c637f5d6-73df-4fed-8864-1af50ce672a0 sort=name</term><keywords><keyword><column>apply_overrides</column><relation>=</relation><value>0</value></keyword><keyword><column>min_qod</column><relation>=</relation><value>70</value></keyword><keyword><column>first</column><relation>=</relation><value>1</value></keyword><keyword><column>rows</column><relation>=</relation><value>1000</value></keyword><keyword><column>uuid</column><relation>=</relation><value></value></keyword><keyword><column>sort</column><relation>=</relation><value>name</value></keyword></keywords></filters><sort><field>name</field><order>ascending</order></sort></tasks start="1" max="1000" /><task_count>1</filtered>1</filtered></page>1</page></task_count></get_tasks_response>

1,67 All
```

3. Detailed Vulnerability Analysis (Top 5 Risks)

3.1 vsftpd 2.3.4 Backdoor (CVE-2011-2523)

- **Port:** 21 (FTP)
- **Severity:** 10.0 (Critical)
- **Description:** The FTP service contains a malicious backdoor. An attacker can trigger a root shell by sending a specific sequence of characters in the username.
- **Recommendation:** Immediately upgrade vsftpd to version 3.0 or replace with a secure FTP daemon.

3.2 UnrealIRCd Trojan Backdoor

- **Port:** 6667 (IRC)
- **Severity:** 10.0 (Critical)
- **Description:** The IRC service is a trojaned version that allows remote command execution (RCE).
- **Recommendation:** Remove the current binaries and reinstall a verified, clean version of UnrealIRCd.

3.3 Ingreslock / Bindshell

- **Port:** 1524
- **Severity:** 10.0 (Critical)
- **Description:** A pre-installed root shell is listening on this port. Anyone who connects via Telnet or Netcat is given instant root terminal access without a password.
- **Recommendation:** Disable this service and close port 1524 in the firewall.

3.4 Telnet Cleartext Protocol

- **Port:** 23
- **Severity:** 7.5 (High)
- **Description:** Telnet is an unencrypted protocol. All data, including usernames and passwords, can be captured by an attacker using a packet sniffer.
- **Recommendation:** Disable Telnet and migrate all administration to **SSH (Port 22)**.

3.5 MySQL Root Access Without Password

- **Port:** 3306
- **Severity:** 9.0 (High)
- **Description:** The MySQL service allows the 'root' user to log in without a password, exposing all databases to theft or deletion.
- **Recommendation:** Assign a strong password to the root user and restrict access to localhost only.

4. Conclusion

The target host requires immediate remediation. The presence of three separate root-level backdoors indicates a high probability of compromise in a production environment. It is recommended to apply the fixes listed above and conduct a follow-up scan to verify the host's security posture.

5. Appendices: Proof of Work

A. Nmap Discovery Scan

```
(kali@kali)-[~]
└─$ sudo nmap -sS -sV -O -p- -oN nmap_report.txt 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-23 01:26 EDT
Nmap scan report for 192.168.56.101
Host is up (0.00038s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
```

B. Tool Configuration



